Reacting acids with carbonates

Carbonates are substances that contain carbon and oxygen. When carbonates are mixed with **acids** they react. They fizz when combined because carbon dioxide is produced. Two other products are also produced: **a salt** and **water**. The general word equation for the reaction between a carbonate and an acid is:

acid + carbonate ⟶ salt + water + carbon dioxide

We know that a salt is formed because if you evaporate the liquid that is left at the end of the reaction, a substance is left behind. The name of the salt that forms depends on the type of carbonate and the type of acid that react. The first part of the salt's name comes from the carbonate and the second part comes from the acid used.

☞ **1** Cross out the incorrect answers.

A reaction between a carbonate and an acid produces	hydrogen gas. carbon dioxide gas.
The first part of the salt's name comes from the	acid. carbonate.
The second part of the salt's name comes from the	acid. carbonate.

2 Name the salt produced when the following reactions occur. The first one has been done for you.

a magnesium carbonate + hydrochloric acid magnesium chloride

b zinc carbonate + hydrochloric acid _____

c copper carbonate + nitric acid _____

d calcium carbonate + nitric acid _____

e lead carbonate + sulfuric acid _____

f copper carbonate + sulfuric acid _____

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